



Use dbt and DuckDB instead of Databricks

Building architectures that fit your needs instead of your vendors

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21/11/2025

Lead data engineer & partner @Dataminded

Early adopter and contributor to DuckDB

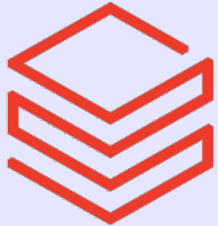
5+ years designing data platforms



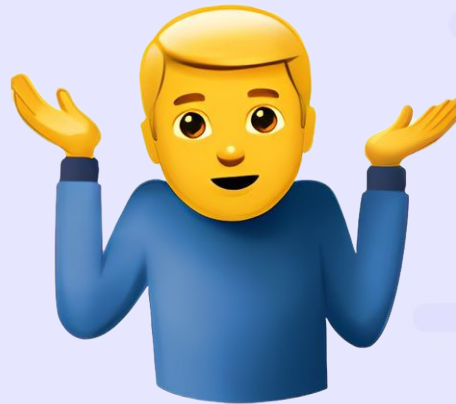
www.linkedin.com/in/nielsclaeys/
medium.com/@niels.claeys



What is your data platform architecture?



databricks



Google
Big Query



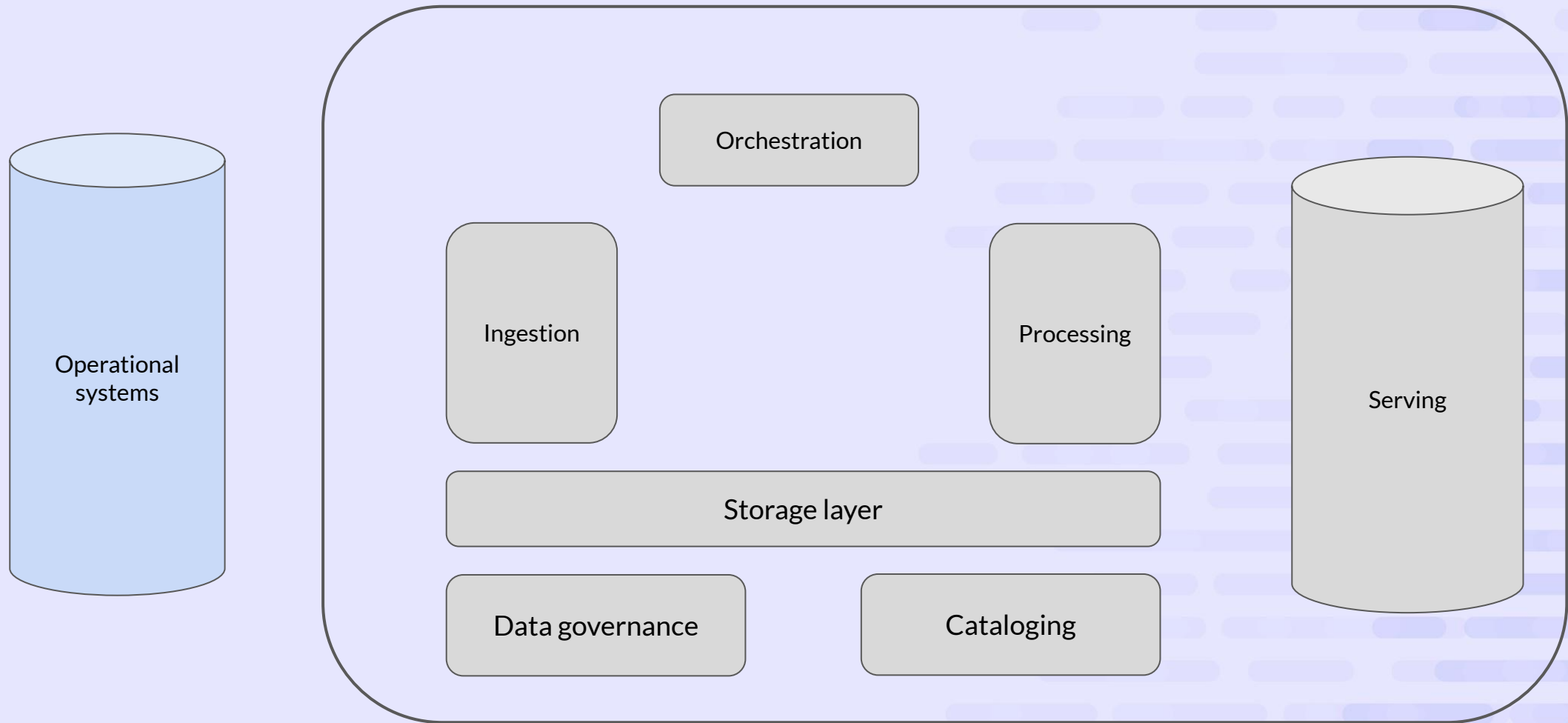
snowflake®



“Architecture is a hypothesis about the future that holds that subsequent change will be confined to that part of the design space encompassed by that architecture.”

Brian Foote







Databricks



Promise

PLATFORM

The Databricks Data Intelligence Platform

Databricks brings AI to your data to help you bring AI to the world.

Unified. Open. Scalable.

The [Databricks Data Intelligence Platform](#) is built on lakehouse architecture, which combines the best elements of data lakes and data warehouses to help you reduce costs and deliver on your data and AI initiatives faster.

Built on open source and open standards, a lakehouse simplifies your data estate by eliminating the silos that historically complicate data and AI.

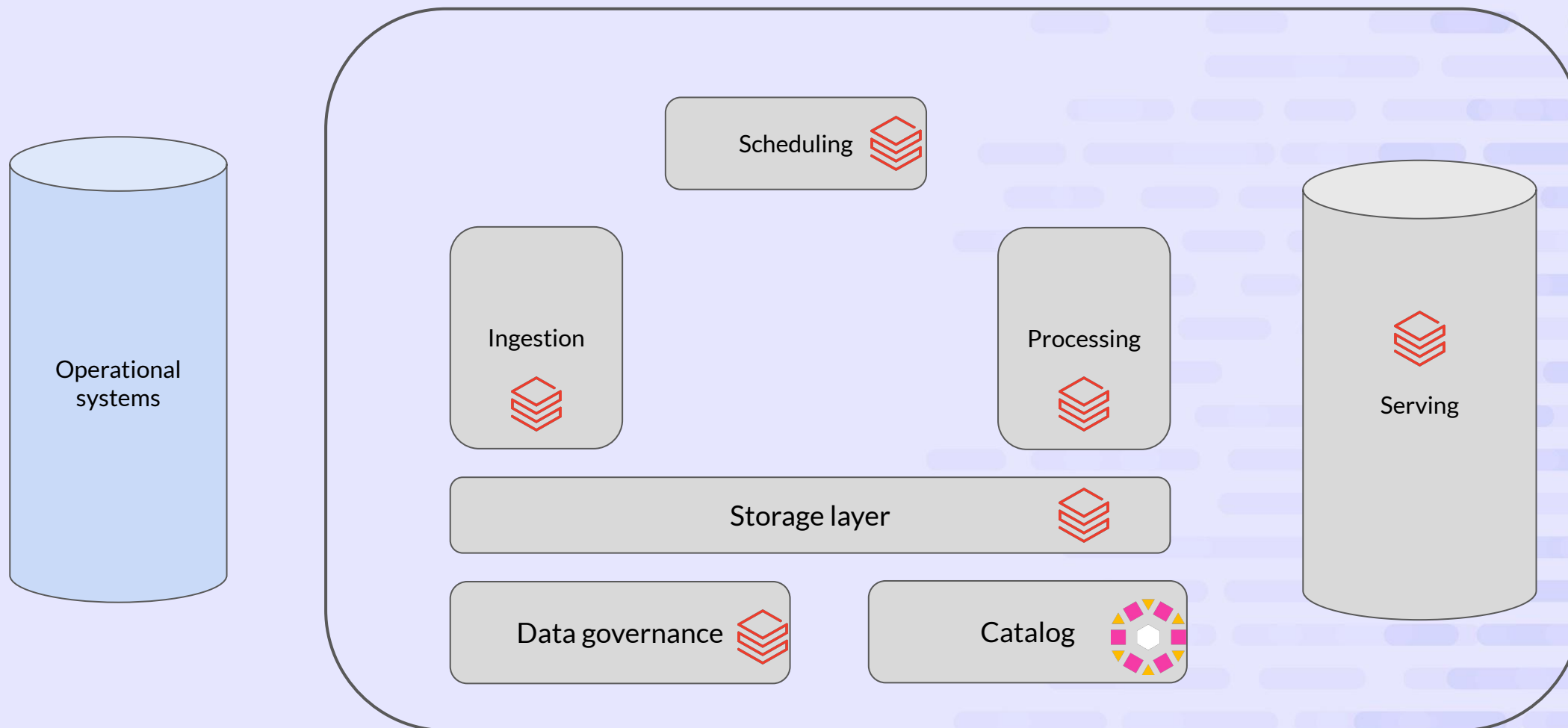
Build your startup on
Databricks

The most powerful platform for data and AI

Apply now

Data Lakehouse Architecture

Unify your data, analytics and AI



Let's get started

- DBX manages all components for you
- Scales from Gb up to Pb of data
- Fast Spark
- 1 tool = Simple

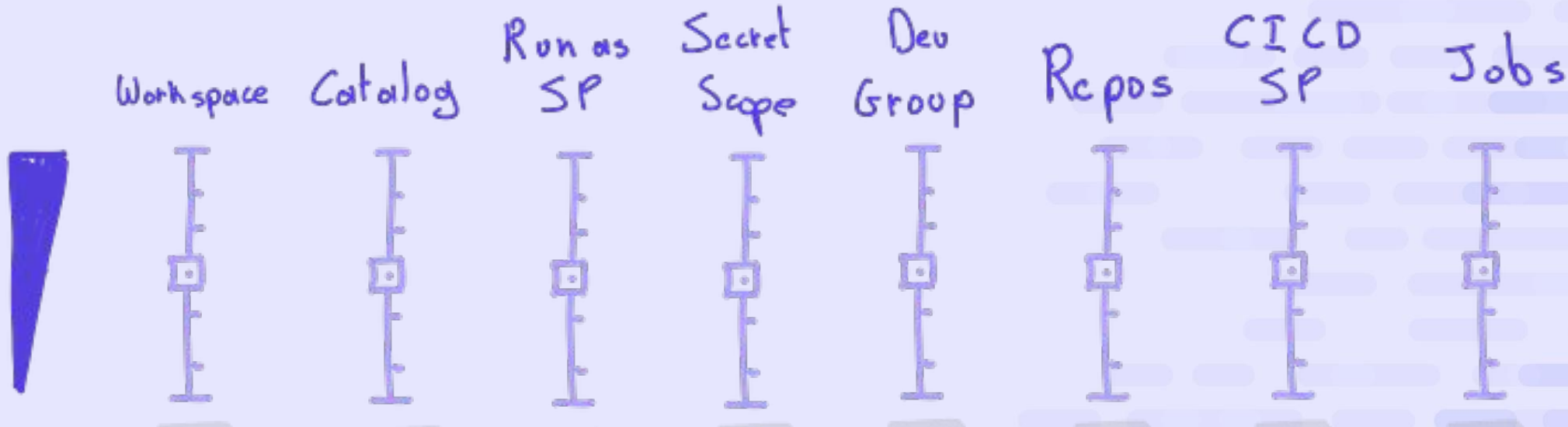


Databricks (hidden) complexity

- Which type of cluster to use?
- How many clusters do I need? (team, domain, environment)
- Which tech stack to run on DBX?
- Managed vs external storage?
- ...



Databricks (hidden) complexity



Getting Data Done

Following

How to set up Databricks for data products?



Stefan Vanraemdonck

Follow

8 min read · Oct 16, 2025



55



medium.com/datamindedbe/how-to-set-up-databricks-for-data-products-7325b99bcaed



Efficiency

- Spark overkill for 80% of pipelines
- dbt on Databricks



Bill shock

- Forgot to shutdown the cluster
- No right-sizing of clusters
- Nodes are expensive



Databricks
(All purpose)

Dsv5 series								
Instance	vCPU(s)	RAM	DBU Count	DBU Price	Pay As You Go Total Price	1 Year Savings Plan (% Savings) Total Price	3 Year Savings Plan (% Savings) Total Price	Spot (% Savings) Total Price
D4s v5	4	16.00 GiB	1.00	€344.222/month	€488.169/month	€453.622/month ~7% savings	€423.393/month ~13% savings	€373.083/month ~24% savings
D8s v5	8	32.00 GiB	2.00	€688.444/month	€976.338/month	€907.243/month ~7% savings	€846.785/month ~13% savings	€746.166/month ~24% savings
D16s v5	16	64.00 GiB	4.00	€1,376.887/month	€1,952.675/month	€1,814.486/month ~7% savings	€1,693.577/month ~13% savings	€1,492.332/month ~24% savings

x3.5

x9

Azure VMs

D4s v5	4	16 GiB	0 GiB	€133.9335/month	€100.9320/month ~24% savings	€69.401/month ~48% savings	€39.9391/month ~70% savings
D8s v5	8	32 GiB	0 GiB	€267.8669/month	€201.8640/month ~24% savings	€138.8089/month ~48% savings	€79.8782/month ~70% savings
D16s v5	16	64 GiB	0 GiB	€535.7339/month	€403.7281/month ~24% savings	€277.6178/month ~48% savings	€159.7557/month ~70% savings
D32s v5	32	128 GiB	0 GiB	€1,071.4678/month	€807.4561/month ~24% savings	€555.2356/month ~48% savings	€319.5114/month ~70% savings

Pros and Cons

- + Scalability
- + Performance
- + Operability

- Evolvability
- Cost efficiency
- Simplicity





dbt and DuckDB



Duckdb

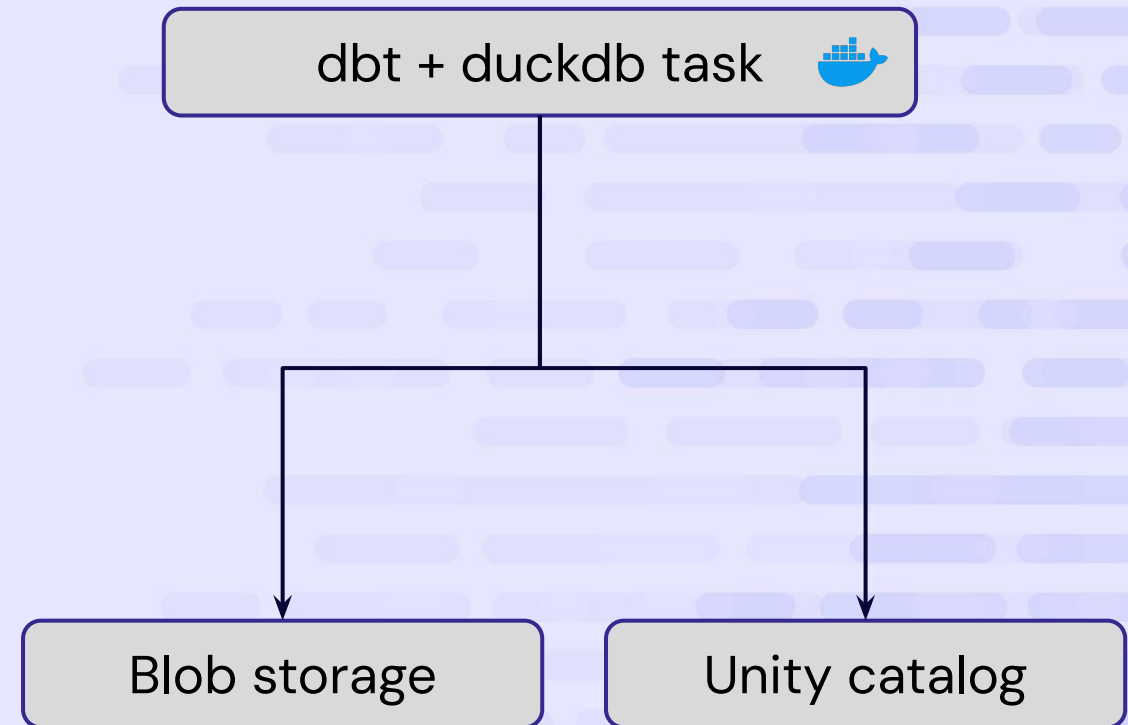
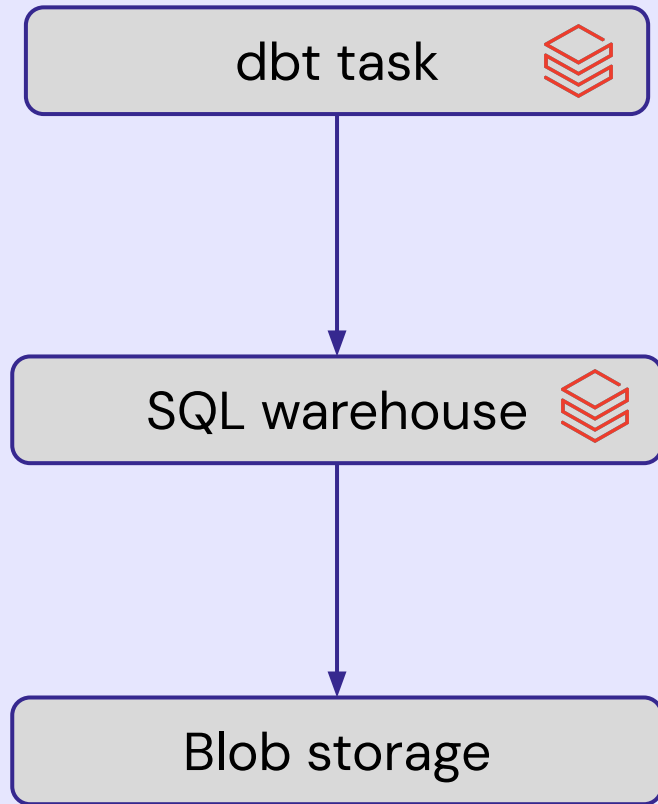
=> Sqlite for analytics

=> DuckDB is an in-process SQL OLAP database management system

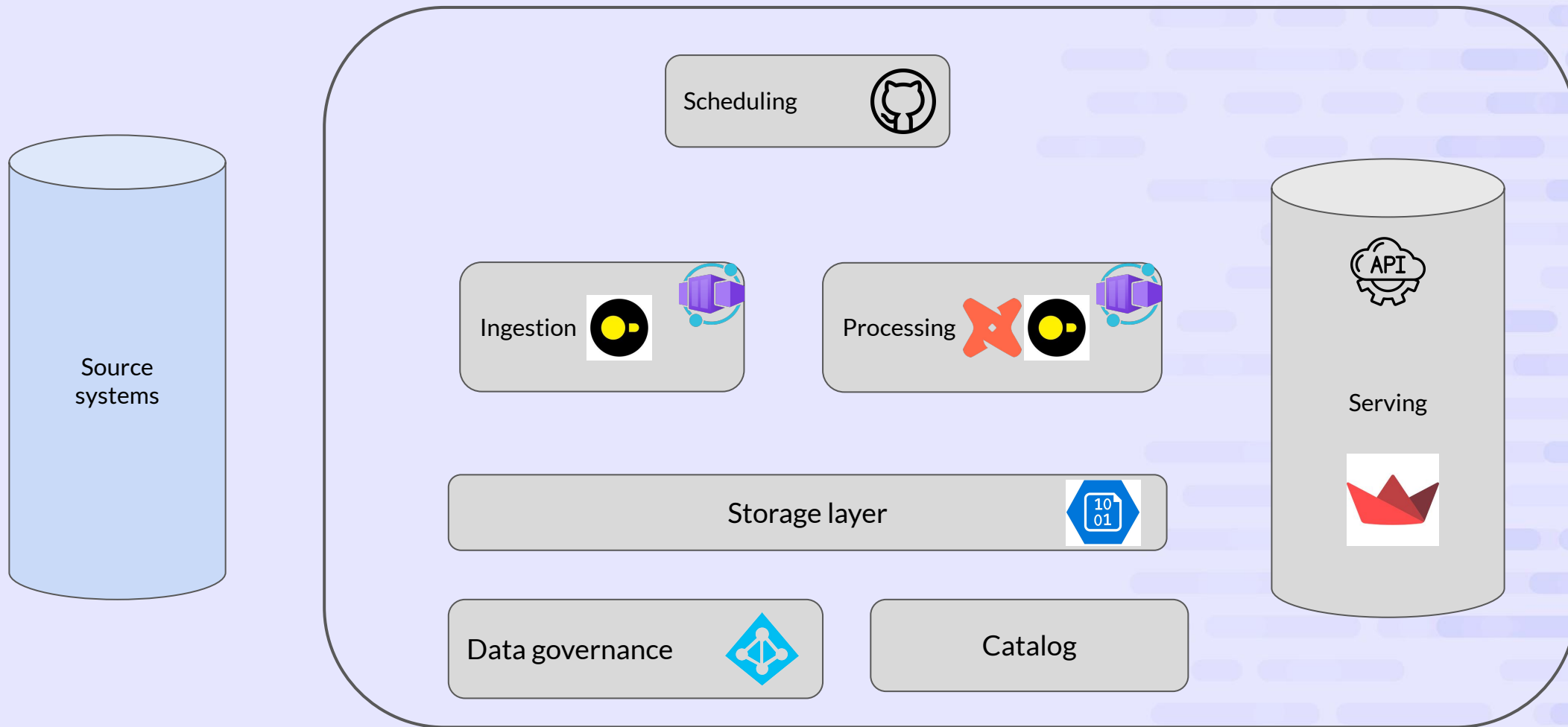
- Easy to use
- Versatility
- Fast: columnar vectorized query engine for OLAP
- Can handle datasets larger than memory



Different processes for running dbt



dbt + DuckDB architecture



Pros and Cons

- + Evolvability
- + Cost efficiency
- + Performance
- + Simplicity

- Scalability
- Operability

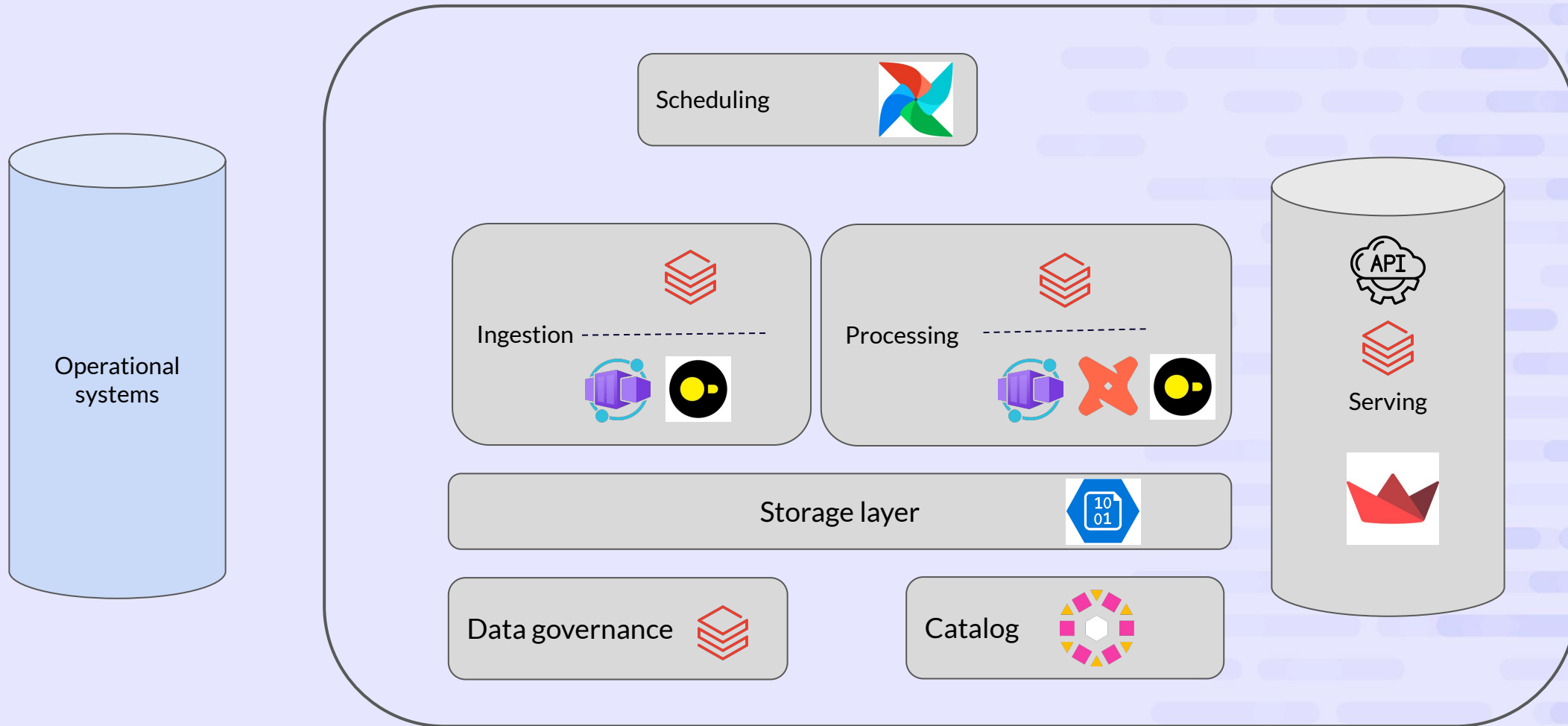




Hybrid setup



Hybrid architecture



Unity catalog integrations

- DuckDB limited support writing delta tables
- Databricks only supports reading Iceberg on delta tables
- Iceberg format is your best bet

<https://docs.databricks.com/aws/en/external-access/iceberg>

Requirements

Databricks supports Iceberg REST catalog access to tables as part of Unity Catalog. You must have Unity Catalog enabled in your workspace to use these endpoints. The following table types are accessible via the Iceberg REST Catalog:

Topic	Read	Write
Managed Iceberg	Yes	Yes
Foreign Iceberg	Yes	No
Managed Delta (with Iceberg reads enabled)	Yes	No
External Delta (with Iceberg read enabled)	Yes	No

Pros and Cons

- + Scalability
- + Performance
- + Performance
- + Evolvability
- + Cost efficiency

- Operability
- Simplicity



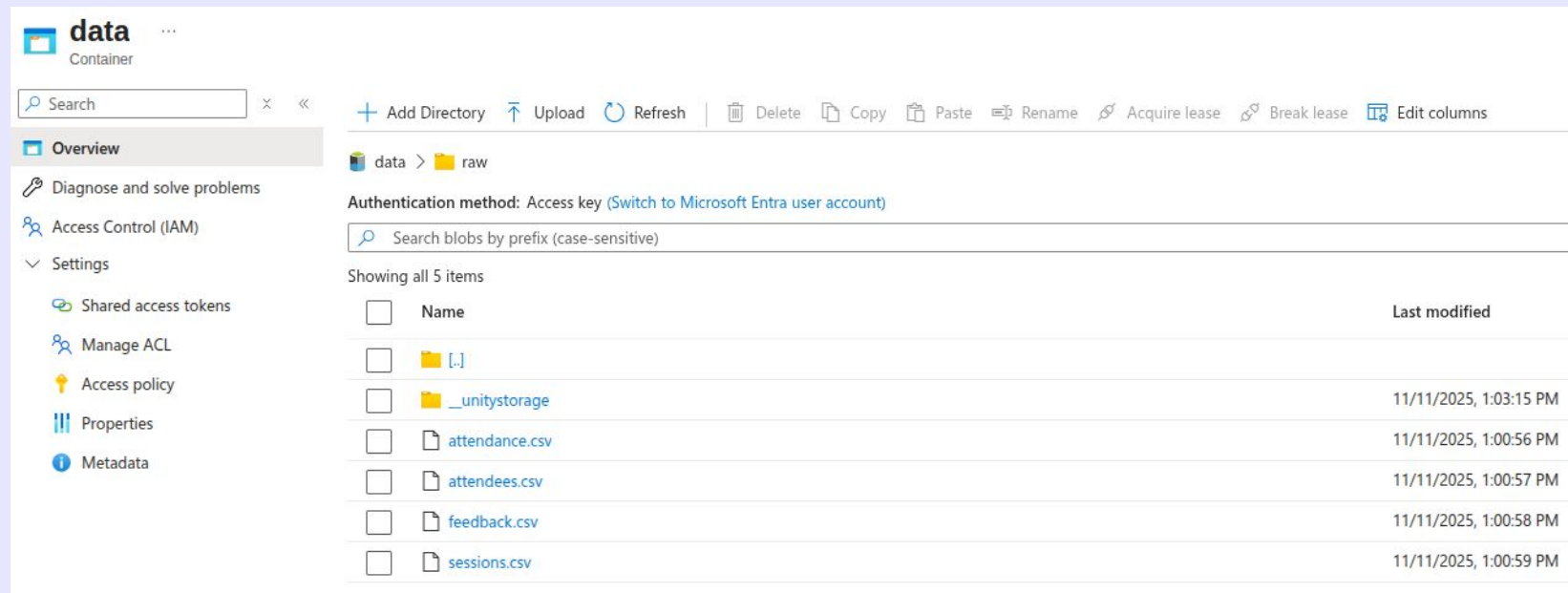


Case study



Case study

- Analyze conference data
- Compare both setups
- Evaluation criteria:
 - Operability
 - Cost
 - Scalability

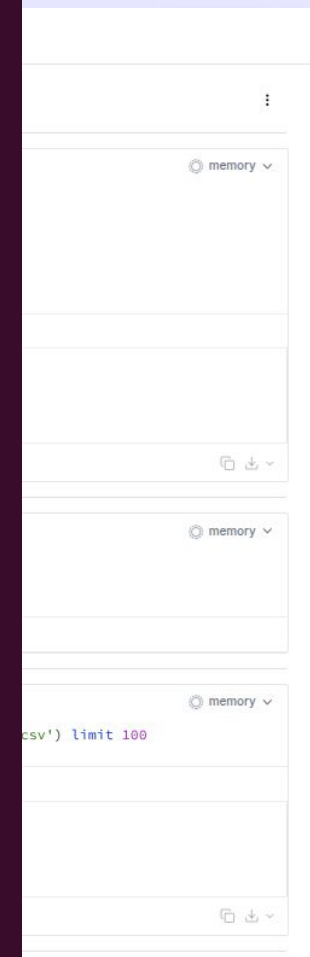


Experimentation: duckdb

```
> dbt run --target=azure
19:23:49 Running with dbt=1.10.11
19:23:50 Registered adapter: duckdb=1.10.0
19:23:50 Unable to do partial parsing because a project config has changed
19:23:51 Found 7 models, 4 sources, 443 macros
19:23:51
19:23:51 Concurrency: 1 threads (target='azure')
19:23:51
19:23:55 1 of 7 START sql external model main.stg_attendance ..... [RUN]
19:23:56 1 of 7 OK created sql external model main.stg_attendance ..... [OK in 1.74s]
19:23:56 2 of 7 START sql external model main.stg_attendees ..... [RUN]
19:23:57 2 of 7 OK created sql external model main.stg_attendees ..... [OK in 1.10s]
19:23:57 3 of 7 START sql external model main.stg_feedback ..... [RUN]
19:23:59 3 of 7 OK created sql external model main.stg_feedback ..... [OK in 1.29s]
19:23:59 4 of 7 START sql external model main.stg_sessions ..... [RUN]
19:24:00 4 of 7 OK created sql external model main.stg_sessions ..... [OK in 1.10s]
19:24:00 5 of 7 START sql external model main.int_attendance_metrics ..... [RUN]
19:24:01 5 of 7 OK created sql external model main.int_attendance_metrics ..... [OK in 1.02s]
19:24:01 6 of 7 START sql external model main.fct_feedback_scores ..... [RUN]
19:24:03 6 of 7 OK created sql external model main.fct_feedback_scores ..... [OK in 1.67s]
19:24:03 7 of 7 START sql external model main.fct_top_sessions_attended ..... [RUN]
19:24:04 7 of 7 OK created sql external model main.fct_top_sessions_attended ..... [OK in 1.56s]
19:24:04
19:24:04 Finished running 7 external models in 0 hours 0 minutes and 12.90 seconds (12.90s).
19:24:04 Completed successfully
19:24:04 Done. PASS=7 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=7
```

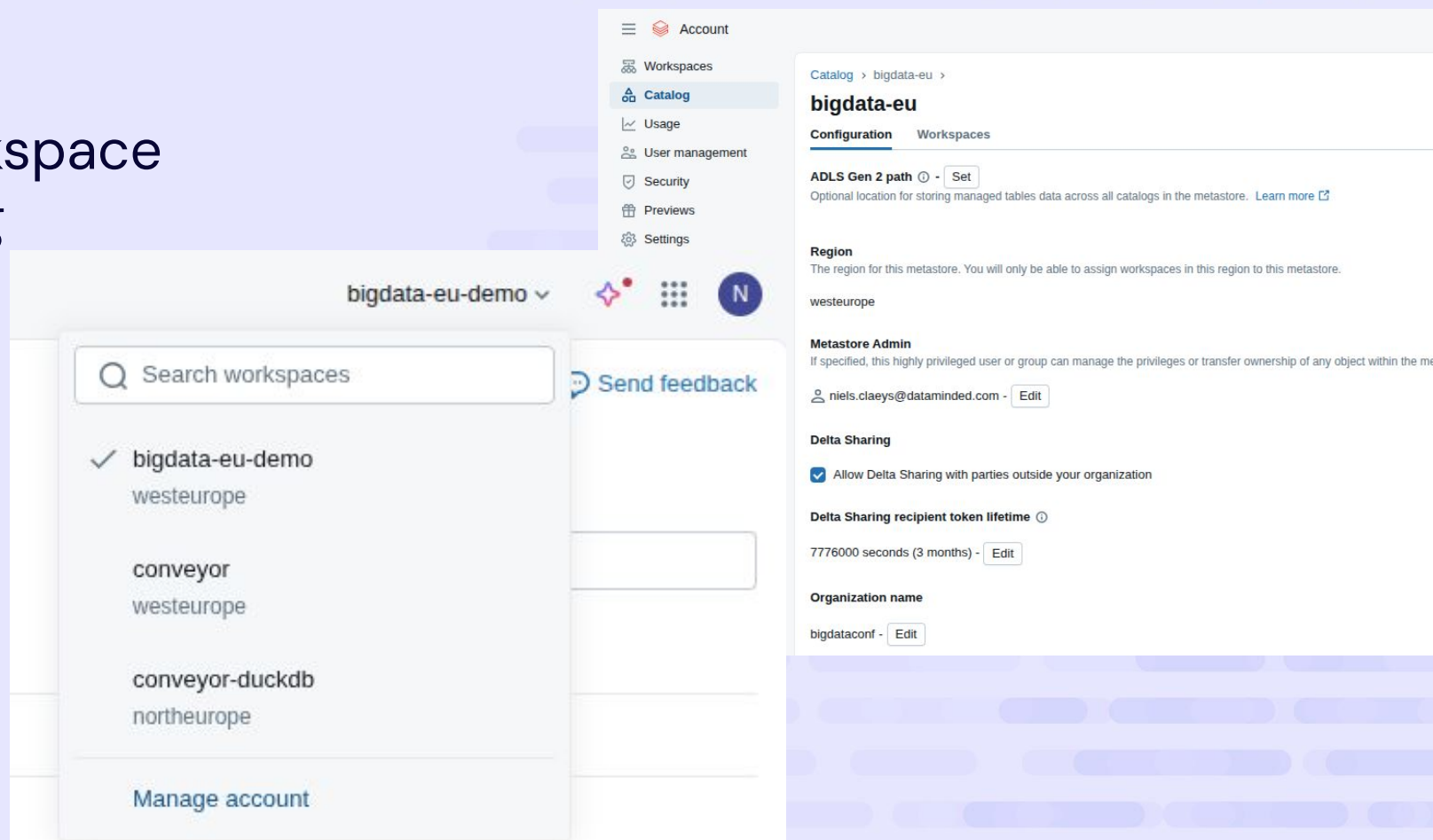


```
default:
azure:
  type:
  path:
  threads:
  extensions:
  - h
  - p
files:
  - f
a
account_name: "databricksbtgdataconf"
account_key: "{{ env_var('DBT_ENV_SECRET_STORAGE_ACCOUNT_KEY') }}"
external_root: "abfs://data/duckdb"
```



Experimentation: databricks

- Create databricks workspace
- Configure unity catalog



<https://learn.microsoft.com/en-us/azure/databricks/data-governance/unity-catalog/create-metastore>
<https://learn.microsoft.com/en-us/azure/databricks/connect/unity-catalog/cloud-storage/external-locations>



Experimentation: databricks

Microsoft Azure | databricks | bigdata-eu-demo

Search: type:notebook

+ New

Workspace

Catalog

Type to search...

For you All

- My organization
 - system
 - bigdataconf
 - default
 - dev_mart
 - dev_staging
 - information_schema
 - raw
 - attendance
 - attendees
 - feedback
 - sessions
 - Delta Shares Received
 - samples
 - Legacy
 - hive_metastore

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

Untitled Notebook 2025-11-15 20:28:53

File Edit View Run Help Python Tabs: ON Last edit was 1 minute ago

Run all Connected Schedule

1 minute ago (30s)

%sql

```
select * from bigdataconf.raw.attendance;
```

Hide performance (1)

View all in query history

Statement	Started At	Tasks	Duration	Rows read	Bytes read	Bytes written
> select * from bigdataconf.raw.attendance	Nov 15, 2025, 08:29 PM	1/1 completed	4 s 193 ms	12,288	4.99 KB	0 B

_sqldf: pyspark.sql.connect.dataframe.DataFrame = [attendee_id: string, session_id: string ... 2 more fields]

Table

	attendee_id	session_id	start_time	_rescued_data
1	ATT00001	5DC0AXXL	2025-11-20T14:10:00.000+00:00	null
2	ATT00002	5DC0AXXL	2025-11-20T14:10:00.000+00:00	null
3	ATT00003	5DC0AXXL	2025-11-20T14:10:00.000+00:00	null
4	ATT00004	5DC0AXXL	2025-11-20T14:10:00.000+00:00	null
5	ATT00005	5DC0AXXL	2025-11-20T14:10:00.000+00:00	null
6	ATT00006	5DC0AXXL	2025-11-20T14:10:00.000+00:00	null
7	ATT00007	5DC0AXXL	2025-11-20T14:10:00.000+00:00	null
9	ATT00009	5DC0AXXL	2025-11-20T14:10:00.000+00:00	null



Productionize: duckdb



```
> tree -L 2
```

```
.
├── analyse
├── dbt_pro
├── Dockerf
├── entryp
├── macros
├── models
│   ├── mar
│   └── sta
├── profile
├── require
├── require
├── seeds
├── snapsh
└── tests
```



```
resource "azurerm_container_app_environment" "dbt-environment" {
  name                = "dbt-dev-environment"
  location             = local.location
  resource_group_name = local.rg_name
  logs_destination    = "azure-monitor"
}

resource "azurerm_container_app_job" "conf-job" {
  name                = "dbt-conf-job"
  location            = local.location
  resource_group_name = local.rg_name
  container_app_environment_id = azurerm_container_app_environment.dbt-environment.id
  identity {
    type = "SystemAssigned"
  }
  replica_timeout_in_seconds = 600
  schedule_trigger_config {
    cron_expression = "0 1 * * *"
    parallelism     = 1
  }
}

template {
  container {
    image = "nilli9990/demo-dbt-conf-analysis:v2"
    name  = "dbt"
    cpu   = 1
    memory = "2Gi"
    env {
      name = "DBT_ENV_SECRET_STORAGE_ACCOUNT_KEY"
      secret_name = "storage-account-key"
    }
  }
}
```

ed/dbt:v1.10.11

irements.txt

at the dbt cache is populated
duce the startup latency

.py"]

Productionize: duckdb

dbt-c
Container A

Search

Overview

Activity log

Access contr

Tags

Diagnose an

Resource vist

Settings

Configur

Logs
dbt-logs-workspace

New Query 1* ... × +

▶ Run

Time range : Last hour

Show : 500000 results

```
1 ContainerAppConsoleLogs
2 | where ContainerGroupName startswith "dbt-conf-job-0on3jsd"
3 | order by TimeGenerated desc
4 | project TimeGenerated, Log, ContainerImage, JobName
```

Results

Chart

TimeGenerated [UTC]	Log	ContainerImage	JobName
> 11/16/2025, 12:36:36.403 PM	INFO:root:dbt run completed successfully.	docker.io/nilli9990/demo-dbt-c...	dbt-conf-job
> 11/16/2025, 12:36:35.878 PM	[0m12:36:35 Done. PASS=7 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=7	docker.io/nilli9990/demo-dbt-c...	dbt-conf-job
> 11/16/2025, 12:36:35.878 PM	[0m12:36:35	docker.io/nilli9990/demo-dbt-c...	dbt-conf-job
> 11/16/2025, 12:36:35.877 PM	[0m12:36:35 [32mCompleted successfully[0m	docker.io/nilli9990/demo-dbt-c...	dbt-conf-job
> 11/16/2025, 12:36:35.877 PM	[0m12:36:35	docker.io/nilli9990/demo-dbt-c...	dbt-conf-job
> 11/16/2025, 12:36:35.843 PM	[0m12:36:35 Finished running 7 external models in 0 hours 0 minutes and 5.85 seconds (5.85s).	docker.io/nilli9990/demo-dbt-c...	dbt-conf-job
> 11/16/2025, 12:36:35.842 PM	[0m12:36:35	docker.io/nilli9990/demo-dbt-c...	dbt-conf-job

logs

onsole / System

onsole / System

Productionize: databricks

A screenshot of the Databricks web interface. The top navigation bar shows the Microsoft Azure logo, the Databricks logo, a search bar with the text "Search data, notebooks, recents, and more...", and a "CTRL + P" shortcut. Below the navigation bar, on the left, is a sidebar with a "New" button (highlighted in red) and links to "Workspace" and "Recents". The main content area displays a breadcrumb trail: "Jobs & Pipelines > devsample_dbt_job > Run 1101984614874061 >". Below this, the text "transform run" is followed by a green "Succeeded" status indicator and a "Lakeflow Jobs UI: OFF" dropdown menu. On the right side of the main content area, there are two buttons: "Edit task" and "Repair run".

Microsoft Azure

databricks

New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Features

Models

Serving

Search data, notebooks, recents, and more...

CTRL + P

bigdata-eu-demo

Jobs & Pipelines

Create new

Ingestion pipeline

Ingest data from apps, databases and files

ETL pipeline

Build ETL pipelines using SQL and Python

Job

Orchestrate notebooks, pipelines, queries and more

Jobs & pipelines

Job runs

Filter by name or ID substring

AllJobsPipelines

Owned by me

Accessible by me

Favorites

Tags

Run as

Create

Name	Type	Tags	Run as	Trigger	Recent runs
devsample_dbt_job	Job		niels.claeys@dataminded.com	Scheduled	

20:52:065 of 7 OK created sql table model dev_mart.int_attendance_metrics [OK in 3.29s]

20:52:076 of 7 START sql table model dev_mart.fct_feedback_scores [RUN]

20:52:096 of 7 OK created sql table model dev_mart.fct_feedback_scores [OK in 2.27s]

20:52:097 of 7 START sql table model dev_mart.fct_top_sessions_attended [RUN]

20:52:117 of 7 OK created sql table model dev_mart.fct_top_sessions_attended [OK in 2.03s]

20:52:11Finished running 7 table models in 0 hours 0 minutes and 27.59 seconds (27.59s).

20:52:11Completed successfully

20:52:11Done. PASS=7 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=7

dbt

Project

Commands

/Workspace/Users/niels.claeys@dataminded.com/.bundle/dbt/dev/files/databricks_conf_analys

1. dbt deps

Cost comparison

- Runtime: daily task of 6 minutes using the smallest on-demand node

# tasks	Databricks *	dbt + DuckDB *2
1	3,47 €/month	0,35 €/month
100	347 €/month	35 €/month
1000	3470 €/month	350 €/month

- Initial setup of dbt + DuckDB is more work
- Maintenance effort is similar both cases

*use job compute (dbt) and SQL serverless (2x small) for warehouse

*2: use 4vcpu and 16Gb RAM on azure container jobs



Operational cost comparison

- Initial setup of dbt + DuckDB is more work (x4)
- Maintenance effort is similar both cases



Scalability

- Databricks is a fit if data > 100Gb
- Both setups favor basic workflows definitions
- Databricks is better when amount datasets increase





Closing thoughts

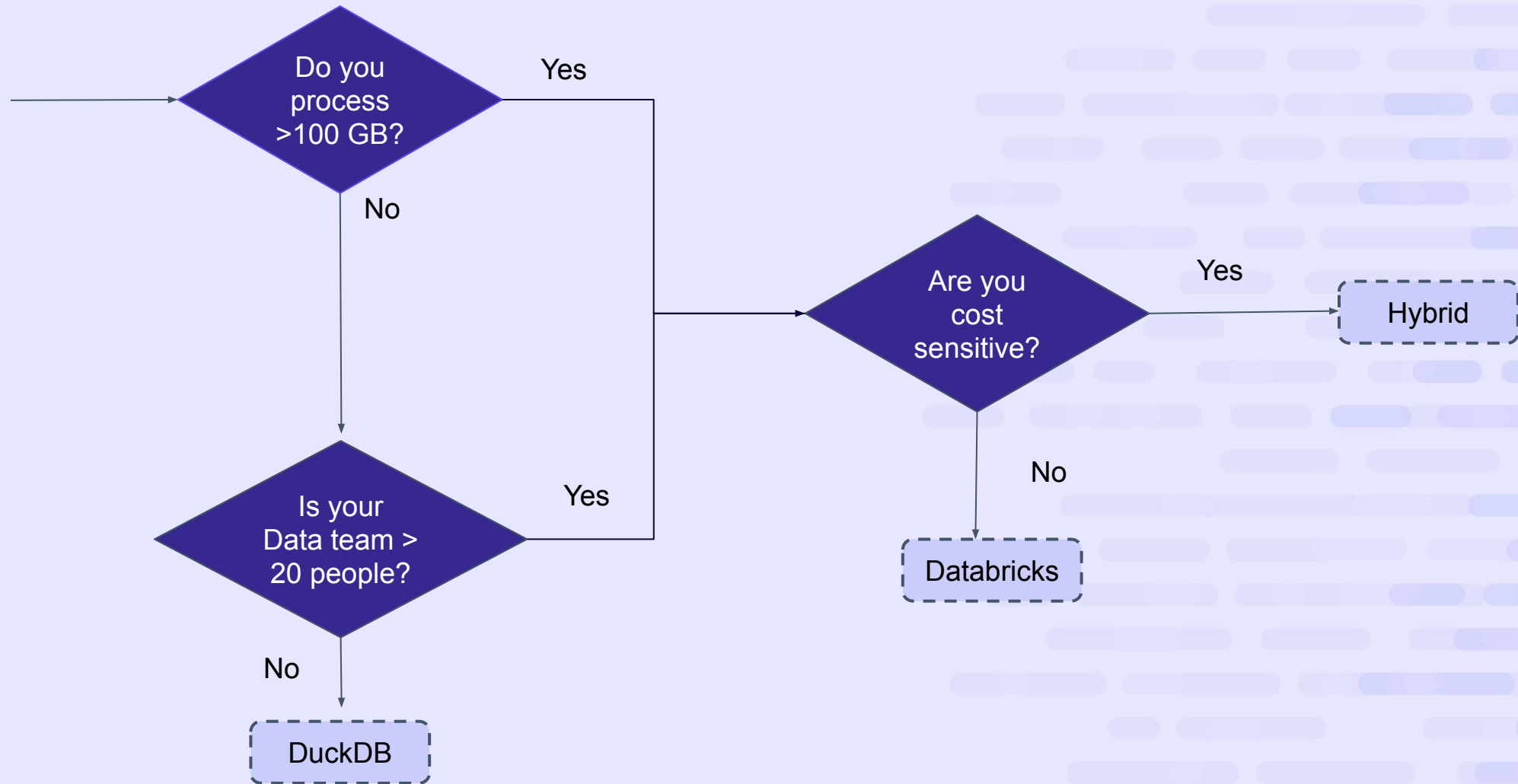


Create an architecture that fits your needs

- Which components do you require
- Key requirements
 - Scalability
 - Performance
 - Operability
 - Evolvability
 - Simplicity
 - Cost efficiency
- There is always a tradeoff between different requirements



Decision framework



How to start

- No redesign entire platform
- Try out DuckDB one use case
github.com/nclaeys/demo-bigdata-europe-talk



Resources

- medium.com/datamindedbe/quacking-queries-in-the-azure-cloud-with-duckdb-14be50f6e141
- medium.com/datamindedbe/use-dbt-and-duckdb-instead-of-spark-in-data-pipelines-9063a31ea2b5
- medium.com/datamindedbe/head-to-head-comparison-of-dbt-sql-engines-497d71535881





Thank you

See you next time! 🙌

